Final Project for Advanced Mathematics

Investigation of Convective Heat Transfer with a Prescribed Velocity Field

An Example of the Advection–Diffusion Equation with Complex Boundary Conditions

Student

Hamid Motallebi ( )

Instructor

Dr. Monjezi

Submission Date

February 2026

Faculty of Mechanical Engineering

College of Engineering, University of Tehran

First Semester, Academic Year 2025–2026

Table of Contents

Introduction **4**

Problem Definition **5**

1-1 Problem Definition and Mathematical Formulation 5

1-2 Physical Assumptions 5

1-3 Governing Equations and Velocity Profile 5

1-4 Boundary and Initial Conditions 9

1-5 Nondimensionalization of the Equations 9

2 Solution of the Problem **11**

2-1 Analytical Solution by Separation of Variables 11

2-1-1 Solution of the Axial Part *11*

2-1-2 Solution of the Transverse Part: Sturm–Liouville Problem *11*

2-1-3 Characteristic Equation and Determination of the Eigenvalues *12*

2-1-4 Final Series Solution *12*

3 Numerical Implementation and Validation **14**

3-1 Computational Grid and Discretization 14

3-2 Formation of the Algebraic System of Equations 14

3-3 Application of Boundary Conditions 14

3-3-1 Symmetry Boundary *14*

3-3-2 Wall Boundary *15*

3-4 Matrix Structure 15

4 Simulation-Code Results (MATLAB) **16**

4-1 Numerical Implementation Using the Implicit Finite-Difference Method 16

4-1-1 Discretization of the Governing Equation *16*

4-1-2 Assembly of the Coefficient Matrix *16*

4-1-3 System Solution and Time Marching *17*

4-1-4 Analysis of Results and Plots *17*

4-2 Part Two: Validation of the Results 20

4-2-1 Part One: Numerical Solution *20*

4-2-2 Part Two: Analytical Solution *20*

4-2-3 Part Three: Validation *21*

4-2-4 Helper Functions at the End of the Code *21*

5 Appendix 1: MATLAB Code Implemented for the Numerical Solution **22**

6 Appendix 2: MATLAB Code Implemented for Validation of the Numerical Results against the Analytical Solution **24**

List of Figures

Figure 1: Color map of temperature over the two-dimensional domain. 18
Figure 2: Temperature variation with radius at different axial locations. 18
Figure 3: Variations of centerline and wall temperature. 19
Figure 4: Output of the second part of the MATLAB code for validating the numerical results against the analytical solution. 22

Introduction

One of the fundamental challenges in the accurate analysis of high-temperature systems is neglecting the interactions among different heat-transfer mechanisms. In many practical applications, the outer wall of a channel is not merely subjected to a constant heat flux; rather, it interacts with the surroundings through natural convection and thermal radiation. Particularly at high temperatures, the contribution of radiative heat transfer becomes significant and makes the governing boundary equation strongly nonlinear because of the fourth-power dependence on temperature. The presence of a parabolic velocity profile (hydrodynamically fully developed flow), instead of a uniform velocity profile, further increases the mathematical complexity of the problem because the governing differential equation then contains spatially varying coefficients, which makes an analytical solution more difficult.

In this study, we present an exact analytical solution for the coupled convection–conduction heat-transfer problem in flow between parallel plates. The main novelty of this work is the simultaneous application of a parabolic velocity profile and a combined nonlinear boundary condition (including convection and linearized radiation). By using the method of separation of variables and solving the Sturm–Liouville differential equation with variable coefficients, the transient temperature distribution in the fluid is obtained. The ultimate objective is to provide a consistent mathematical formulation capable of predicting the thermal behavior of the system under the influence of the Biot number and combined environmental effects, and to compare and validate the results against a finite-difference numerical solution.

Problem Definition

1-1 Problem Definition and Mathematical Formulation

In this study, heat transfer between two parallel plates is investigated for ($Hydrodynamically\ Fully\ Developed$) flow that is also ($Thermally\ Developing$). The problem geometry consists of a two-dimensional channel with height $2H$ and length L. The x axis is aligned with the fluid-flow direction, and the y axis is normal to the flow and measured from the channel centerline (schematic of the problem).

1-2 Physical Assumptions

The following assumptions are adopted to derive the governing equations:

1. The fluid flow is laminar ($Laminar$), incompressible, and steady.
2. The fluid properties (density ρ, viscosity μ, thermal conductivity k, and heat capacity cp) are assumed constant.
3. The velocity profile is parabolic ($Parabolic$) and is not affected by temperature.
4. Axial heat conduction in the (x) direction is neglected compared with axial convection (boundary-layer assumption or high Peclet number).
5. Viscous dissipation ($Viscous\ Dissipation$) is negligible.
6. Radiative heat transfer occurs at the walls, and the surroundings are treated as a blackbody at temperature $T\infty$.

1-3 Governing Equations and Velocity Profile

1-3-1 General Energy Equation (in Two-Dimensional Cartesian Coordinates)

The general heat-transfer equation for an incompressible fluid with constant properties is written as follows:

$$\rho c_p \left(\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y}\right) = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + \mu\Phi + \dot{q}$$

where:

- **Left-hand side: energy changes due to time variation and convection (**$Convection$).
- **Right-hand side: conductive heat transfer (**$Conduction$), viscous dissipation ($\mu\Phi$), and internal heat generation ($\dot{q}$)

1-3-2 Application of the Problem Assumptions

To obtain the desired equation, the following assumptions are applied:

1- Steady state: temperature does not vary with time.

$$\frac{\partial T}{\partial t} = 0$$

2- Hydrodynamically fully developed flow: the vertical velocity component is zero, and the horizontal velocity depends only on y.

$$v = 0, \quad u = u(y)$$

(Therefore, the term $v\frac{\partial T}{\partial y}$ is eliminated.)

3- No heat generation and negligible viscous dissipation: there is no heat source and no heat

generation due to friction.

$$\dot{q} = 0, \quad \mu\Phi \approx 0$$

4- Neglect of axial conduction:

This is the most important assumption in Graetz-type problems. Heat transfer in the x direction is assumed to occur primarily through fluid motion (convection), while heat conduction in the x direction is negligible compared with convection ($Pe \gg 1$). In the y direction (transverse direction), however, because there is no velocity component, conduction is the dominant mechanism.

$$\left|u\frac{\partial T}{\partial x}\right| \gg \left|k\frac{\partial^2 T}{\partial x^2}\right| \Longrightarrow \frac{\partial^2 T}{\partial x^2} \approx 0$$

1-3-3 Simplified Equation

After applying the above simplifications, the equation reduces to the following form:

$$u(y)\frac{\partial T}{\partial x} = k\frac{\partial^2 T}{\partial y^2}$$

- **Definition of Thermal Diffusivity**

Dividing both sides of the equation by ρc_p gives:

$$u(y)\frac{\partial T}{\partial x} = \frac{k}{\rho c_p}\frac{\partial^2 T}{\partial y^2}$$

The thermal diffusivity (α) is defined as follows:

$$\alpha = \frac{k}{\rho c_p}$$

1-3-4 Velocity Profile for Plane Poiseuille Flow Between Two Plates

1- Navier–Stokes equation in the x direction:

The general momentum-conservation equation for an incompressible fluid with constant viscosity μ in the x direction (the flow direction) is written as follows:

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right) = -\frac{\partial P}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) + \rho g_x$$

2- Application of assumptions for simplification

Consider a channel in which the fluid moves in the x direction, with the plates located at $y = -H$ (lower plate) and $y = +H$ (upper plate). The channel centerline is at $y = 0$. The following assumptions are applied:

- **Steady state: the flow does not vary with time.**

$$\frac{\partial u}{\partial t} = 0$$

- **Fully developed flow: the velocity profile does not change along the channel. Therefore, the velocity in the** x direction is invariant along the channel and depends only on y.

$$\frac{\partial u}{\partial x} = 0 \quad , \quad v = 0,\ w = 0$$

According to the continuity equation $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$, *because the first term is zero,* v *is also zero.*

- **Two-dimensional flow: the channel is very wide in the** z direction; therefore, there are no variations in the z direction.

$$\frac{\partial^2 u}{\partial z^2} = 0$$

- **Neglect of gravity (horizontal channel):**

$$g_x = 0$$

After applying these simplifications, the entire left-hand side of the equation (the inertial terms) becomes zero, and the equation reduces to the following simple form:

$$0 = -\frac{\partial P}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2}$$

Rearranging the equation gives:

$$\mu \frac{d^2 u}{dy^2} = \frac{dP}{dx}$$

(Because u is a function only of y, and P is a function only of x, the partial derivatives ∂ are converted to ordinary derivatives d.)

3- Integration of the equation

In the equation above, the left-hand side is a function only of y, whereas the right-hand side is a function only of x. This equality can hold only if both sides are equal to a constant. Thus, the pressure gradient $\frac{dP}{dx}$ is constant along the channel.

We now integrate both sides of the equation twice with respect to y:

First integration:

$$\frac{du}{dy} = \left(\frac{1}{\mu}\frac{dP}{dx}\right) y + C_1$$

Second integration:

$$u(y) = \left(\frac{1}{2\mu}\frac{dP}{dx}\right) y^2 + C_1 y + C_2$$

We now have an equation containing two constants, (C_1, C_2), which must be determined from the boundary conditions.

4- Application of boundary conditions

- **First boundary condition (symmetry at the channel centerline):**

At the channel centerline $(y = 0)$, the velocity profile reaches its maximum value; therefore, the velocity gradient is zero.

$$\frac{du}{dy} = 0 At\ y = 0$$

Substitution into the result of the first integration gives:

$$0 = \left(\frac{1}{\mu}\frac{dP}{dx}\right)(0) \Longrightarrow C_1 = 0$$

- **Second boundary condition (no-slip condition at the wall):**

The fluid in contact with the wall assumes the wall velocity. Because the walls are stationary, the fluid velocity at the walls $(y = H, y = -H)$ is zero.

$$At\ y = H \implies u = 0$$

Substitution into the result of the second integration (taking $C_1 = 0$) gives:

$$0 = \left(\frac{1}{2\mu}\frac{dP}{dx}\right) H^2 + C_2 \Longrightarrow C_2 = -\left(\frac{1}{2\mu}\frac{dP}{dx}\right) H^2$$

5- Derivation of the final velocity relation

The constants C_1, C_2 are now substituted into the velocity equation:

$$u(y) = \left(\frac{1}{2\mu}\frac{dP}{dx}\right)y^2 - \left(\frac{1}{2\mu}\frac{dP}{dx}\right)H^2$$

Factoring out the common term gives:

$$u(y) = -\left(\frac{1}{2\mu}\frac{dP}{dx}\right)H^2\left[1 - \frac{y^2}{H^2}\right]$$

6- Definition of the maximum velocity (U_0)

The maximum fluid velocity occurs at the channel centerline ($y = 0$). By setting $y = 0$ in the relation above, the maximum velocity ($U0$) is obtained:

$$u(0) = -\left(\frac{1}{2\mu}\frac{dP}{dx}\right)H^2$$

(Note: because pressure decreases along the flow direction, $\frac{dP}{dx}$ is negative. Therefore, the minus sign outside the parentheses makes U_0 positive.)

The entire expression $-\left(\frac{1}{2\mu}\frac{dP}{dx}\right)H^2$ in the original relation is now replaced by U_0:

$$u(y) = U_0\left(1 - \left(\frac{y}{H}\right)^2\right)$$

1-3-5 Final Result

By substituting α, we obtain:

$$u(y)\frac{\partial T}{\partial x} = \alpha\frac{\partial^2 T}{\partial y^2}$$

Physical interpretation of the equation:

The rate of energy change convected by the fluid along the channel (x direction) is equal to the rate at which energy diffuses by conduction across the channel (y direction).

- Left-hand side: convective heat transfer ($Convection$) in the flow direction.
- Right-hand side: conductive heat transfer ($Conduction$) normal to the flow.

where $\alpha = k/(\rho c_p)$ is the thermal diffusivity.

The velocity profile for fully developed flow between two parallel plates was also obtained in the previous section as follows:

$$u(y) = U_{max}\left[1 - \left(\frac{y}{H}\right)^2\right]$$

where U_{max} is the maximum velocity at the channel centerline ($y = 0$).

1-4 Boundary and Initial Conditions

The conditions governing the boundaries of the solution domain are as follows:

1. **Inlet condition: at the channel inlet, the fluid has a uniform temperature T_0:**

$$T(0, y) = T_0$$

2. **Symmetry condition: because of geometric and flow symmetry, the temperature gradient at the centerline is zero:**

$$\left.\frac{\partial T}{\partial y}\right|_{y=0} = 0$$

3. **Wall boundary condition (combined convection–radiation BC): at the channel wall**($y = H$), the conductive heat flux is equal to the sum of the convective and radiative heat fluxes:

$$-k\left.\frac{\partial T}{\partial y}\right|_{y=H} = h_{conv}(T - T_\infty) + \epsilon\sigma(T^4 - T_\infty^4)$$

To make an analytical solution possible, the nonlinear radiation term is linearized about the ambient temperature $T\infty$. By defining an effective radiative heat-transfer coefficient $h_{rad} \approx 4\epsilon\sigma T_{\infty}^{3}$,

the boundary condition is rewritten as follows:

$$-k\frac{\partial T}{\partial y}\Big|_{y=H} = (h_{conv} + h_{rad})(T - T_{\infty}) = h_{eff}(T - T_{\infty})$$

1-5 Nondimensionalization of the Equations

To generalize the results and facilitate the mathematical solution, the following dimensionless parameters are defined:

- Dimensionless temperature: $\theta = \frac{T-T_{\infty}}{T_0-T_{\infty}}$
- Dimensionless transverse coordinate: $\eta = \frac{y}{H}$
- Dimensionless axial coordinate (Graetz scaling): $\xi = \frac{x/H}{Pe} = \frac{x\alpha}{U_{max}H^2}$
- Effective Biot number: $Bi = \frac{h_{eff}H}{k}$

Substituting these variables into the energy equation and the boundary conditions yields the final dimensionless form of the problem to be solved (the Sturm–Liouville equation):

Partial differential equation (PDE):

$$(1-\eta^2)\frac{\partial\theta}{\partial\xi} = \frac{\partial^2\theta}{\partial\eta^2}$$

Dimensionless boundary conditions:

1. At the inlet:

$$\theta(0,\eta) = 1$$

2. At the centerline:

$$\frac{\partial\theta}{\partial\eta}\Big|_{\eta=0} = 0$$

3. At the wall (Robin condition):

$$\frac{\partial\theta}{\partial\eta}\Big|_{\eta=1} + Bi\cdot\theta(\xi,1) = 0$$

Solution of the Problem

2-1 Analytical Solution by Separation of Variables

Our objective is to solve the energy partial differential equation subject to the boundary conditions defined in the previous section. The governing equation was:

$$(1-\eta^2)\frac{\partial\theta}{\partial\xi}=\frac{\partial^2\theta}{\partial\eta^2}$$

To solve this equation, the method of separation of variables is used. We assume that the solution $\theta(\xi,\eta)$ is the product of two independent functions, one depending only on the axial coordinate (ξ) and the other only on the transverse coordinate (η):

$$\theta(\xi,\eta)=X(\xi)\cdot Y(\eta)$$

Substituting this assumption into the original equation and rearranging the terms gives:

$$\frac{1}{X}\frac{dX}{d\xi}=\frac{1}{(1-\eta^2)Y}\frac{d^2Y}{d\eta^2}$$

Because the left-hand side of the equation is a function only ofξ, while the right-hand side is a function only of η, both sides must be equal to a constant. To obtain a bounded physical solution that decays along the channel, this constant is taken to be negative $(-\lambda^2)$:

$$\frac{1}{X}\frac{dX}{d\xi}=-\lambda^2 \quad , \quad \frac{1}{(1-\eta^2)Y}\frac{d^2Y}{d\eta^2}=-\lambda^2$$

This leads to two ordinary differential equations (ODEs):

2-1-1 Solution of the Axial Part (Axial Solution)

The first equation is a simple first-order ODE:

$$\frac{dX}{d\xi}+\lambda^2X=0$$

whose general solution is exponential:

$$X(\xi)=C\cdot e^{-\lambda^2\xi}$$

2-1-2 Solution of the Transverse Part: Sturm–Liouville Problem (The Graetz Problem)

The second equation is an eigenvalue problem known as the Graetz equation:

$$\frac{d^2Y}{d\eta^2}+\lambda^2(1-\eta^2)Y=0$$

This equation is a Sturm–Liouville equation with weight function $w(\eta)=(1-\eta^2)$.

The boundary conditions for $Y(\eta)$ are obtained from the original boundary conditions of the problem:

1. Symmetry condition at $\eta=0$:

$$Y'(0)=0$$

2. Robin condition at $\eta=1$:

$$Y'(1)+Bi\cdot Y(1)=0$$

Because the Graetz equation has variable coefficients (the term $1-\eta^2$), it does not have a simple analytical solution such as a sine or cosine. The standard approach for this equation is the power-series method, also known as the Frobenius method.

Assume that the solution can be expressed as a power series:

$$Y(\eta) = \sum_{n=0}^{\infty} a_n \eta^n$$

Applying the symmetry condition ($Y'(0) = 0$), shows that only even powers of η remain. Therefore, the series is rewritten as follows:

$$Y(\eta) = \sum_{k=0}^{\infty} b_k \eta^{2k} = b_0 + b_1\eta^2 + b_2\eta^4 + \cdots$$

For convenience and initial normalization, let $b_0 = 1$. Substituting this series into the original differential equation and setting the coefficients of different powers of η, equal to zero yields a recurrence relation for the coefficients b_k:

$$b_0 = 1$$

$$b_1 = -\frac{\lambda^2}{2} b_0$$

$$b_k = \frac{\lambda^2 (b_{k-2} - b_{k-1})}{2k(2k-1)} \quad \text{for } k \geq 2$$

This relation allows us to construct the eigenfunction $Y(\eta, \lambda)$ as a function of η and the unknown parameter λ.

2-1-3 Characteristic Equation and Determination of the Eigenvalues

To determine the admissible values of λ (the eigenvalues), the second boundary condition at the wall must be satisfied. Substitution of the obtained series into the Robin boundary condition yields the characteristic equation:

$$\left(\sum_{k=0}^{\infty} 2k\, b_k(\lambda)\right) + Bi \cdot \left(\sum_{k=0}^{\infty} b_k(\lambda)\right) = 0$$

The roots of this transcendental equation are the eigenvalues of the problem (λ_n). These values must be determined numerically (for example, by the Newton–Raphson or Brent method). For each λ_n, there is a corresponding eigenfunction $Y_n(\eta)$.

2-1-4 Final Series Solution

The general solution of the problem is the superposition of all eigenmodes:

$$\theta(\xi, \eta) = \sum_{n=1}^{\infty} C_n Y_n(\eta) e^{-\lambda_n^2 \xi}$$

To determine the unknown coefficients C_n,, we use the inlet initial condition ($\theta(0, \eta) = 1$) and the orthogonality property of the Sturm–Liouville eigenfunctions. For this problem, the orthogonality relation, considering the weight function $w(\eta) = (1 - \eta^2)$, is:

$$\int_0^1 (1-\eta^2) Y_n(\eta) Y_m(\eta)\, d\eta = 0 \quad \text{if } n \neq m$$

Multiplying both sides of the initial condition by $(1 - \eta^2)Y_n(\eta)$ and integrating over the domain [0,1], the coefficient C_n is obtained as follows:

$$C_n = \frac{\int_0^1 (1-\eta^2) \cdot 1 \cdot Y_n(\eta)\, d\eta}{\int_0^1 (1-\eta^2) Y_n^2(\eta)\, d\eta}$$

Numerical Implementation and Validation

Because of the nature of the governing equation and the presence of the variable coefficient $(1-\eta^2)$, which becomes zero at the wall, the numerical solution requires special stability considerations. In this project, an implicit finite-difference method is used.

3-1 Computational Grid and Discretization

The solution domain is divided into a rectangular grid with N nodes in the transverse direction (η) and M steps in the axial direction (ξ):

$$\eta_i = (i-1)\Delta\eta, \quad i = 1, \dots, N$$
$$\xi^j = (j-1)\Delta\xi, \quad j = 1, \dots, M$$

The governing differential equation is:

$$(1-\eta^2)\frac{\partial\theta}{\partial\xi} = \frac{\partial^2\theta}{\partial\eta^2}$$

For discretization, a backward difference $(Backward)$ is used for the temporal (axial) derivative, and a central difference $(Central)$ is used for the spatial (transverse) derivative. This results in the unconditional stability $(Unconditional\ Stability)$ of the method:

$$(1-\eta_i^2)\frac{\theta_i^{j+1}-\theta_i^j}{\Delta\xi} = \frac{\theta_{i+1}^{j+1}-2\theta_i^{j+1}+\theta_{i-1}^{j+1}}{(\Delta\eta)^2}$$

3-2 Formation of the Algebraic System of Equations

By defining the parameter $r = \frac{\Delta\xi}{(\Delta\eta)^2}$ and the velocity profile $u_i = 1-\eta_i^2$, the discretized equation is rearranged so that the unknowns at the new step $(j+1)$ are placed on the left-hand side, while the known quantities from the previous step (j) are placed on the right-hand side:

$$u_i(\theta_i^{j+1}-\theta_i^j) = r(\theta_{i+1}^{j+1}-2\theta_i^{j+1}+\theta_{i-1}^{j+1})$$

After rearranging the terms:

$$\underbrace{-r}_{A_L}\theta_{i-1}^{j+1} + \underbrace{(u_i+2r)}_{A_D}\theta_i^{j+1} + \underbrace{-r}_{A_U}\theta_{i+1}^{j+1} = \underbrace{u_i\theta_i^j}_{RHS}\ AL$$

This relation forms a tridiagonal system of linear equations for the interior nodes $(i = 2 \dots N-1)$.

3-3 Application of Boundary Conditions

3-3-1 Symmetry Boundary ($\eta = 0$, corresponding to node $i = 1$)

The boundary condition is $\frac{\partial\theta}{\partial\eta} = 0$. Using a ghost-node method at $i = 0$:

$$\frac{\theta_2^{j+1}-\theta_0^{j+1}}{2\Delta\eta} = 0 \quad \Rightarrow \quad \theta_0^{j+1} = \theta_2^{j+1}$$

Writing the discretized equation for node $i = 1$ and substituting θ_0 gives:

$$(u_1+2r)\theta_1^{j+1} - 2r\theta_2^{j+1} = u_1\theta_1^j$$

(Note: here, $u_1 = 1-0^2 = 1$.)

3-3-2 Wall Boundary ($\eta = 1$, corresponding to node $i = N$)

At this point, the fluid velocity is zero ($u_N = 0$). If the original equation is written at this point, the left-hand side becomes zero and only the pure- diffusion equation remains. The Robin boundary

condition is:

$$\frac{\partial \theta}{\partial \eta} + Bi \cdot \theta = 0$$

Using a one-sided finite difference (or a central difference with a ghost node) and combining it with the governing equation at the boundary yields the following relation for the last row of the matrix. For greater stability, the boundary condition is imposed directly in discretized form (second-order backward or one-sided difference):

$$\frac{\theta_N - \theta_{N-1}}{\Delta\eta} + Bi \cdot \theta_N = 0$$
$$\Rightarrow \theta_N(1 + Bi\Delta\eta) - \theta_{N-1} = 0$$

This equation forms row N of the matrix.

3-4 Matrix Structure

At each time step, the system of equations $[A]\Theta^{j+1} = B$ is solved, where the matrix A has the following form:

$$\begin{bmatrix} 1+2r & -2r & 0 & \cdots & 0 \\ -r & u_2+2r & -r & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & -r & u_{N-1}+2r & -r \\ 0 & \cdots & 0 & -1 & 1+Bi\times\Delta\eta \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_{N-1} \\ \theta_N \end{bmatrix} = \begin{bmatrix} 1\times\theta_1^j \\ u_2\theta_2^j \\ \vdots \\ u_{N-1}\theta_{N-1}^j \\ 0 \end{bmatrix}$$

SIMULATION-CODE RESULTS (MATLAB)

4-1 Numerical Implementation Using the Implicit Finite-Difference Method (Implicit FDM)

To solve the partial differential equation governing the problem (the energy equation with a parabolic velocity profile), the Implicit Finite Difference Method is used. Because of its unconditional stability, this method is particularly suitable for the small spatial steps (Δη) required near the walls. The details of the MATLAB implementation are described below.

4-1-1 Discretization of the Governing Equation

The dimensionless governing equation is:

$$u(\eta)\frac{\partial \theta}{\partial \xi} = \frac{\partial^2 \theta}{\partial \eta^2}$$

where $u(\eta) = 1 - \eta^2$.

For discretization, a backward difference ($Backward$) is used for the axial derivative(ξ), and a central difference ($Central$) is used for the transverse derivative (η) at the new step ($j+1$):

$$u_i \frac{\theta_i^{j+1} - \theta_i^j}{\Delta\xi} = \frac{\theta_{i+1}^{j+1} - 2\theta_i^{j+1} + \theta_{i-1}^{j+1}}{\Delta\eta^2}$$

By defining the stability parameter $r = \frac{\Delta\xi}{\Delta\eta^2}$, the algebraic equation for the interior points is rearranged into the following form:

$$-r\theta_{i-1}^{j+1} + (u_i + 2r)\theta_i^{j+1} - r\theta_{i+1}^{j+1} = u_i\theta_i^j$$

This relation is implemented in the MATLAB code inside the loop $for\ i = 2{:}N-1$, leading to a linear system of equations of the form $[A]\Theta^{j+1} = B$.

4-1-2 Assembly of the Coefficient Matrix

The matrix A is tridiagonal and contains the coefficients of the unknowns at the new step.

a) Interior points

For nodes $i = 2$ through $N-1$, the matrix elements are defined as follows:

- Main diagonal: $A(i,i) = u(i) + 2 * r$
- Upper and lower diagonals: $A(i,i-1) = -r$ and $A(i,i+1) = -r$
- Right-hand-side vector (B): $B(i) = u(i) * Theta(i,j)$ (containing the temperature values from the previous step).

b) Symmetry boundary condition at the center ($Symmetry\ at\ \eta = 0$)

At the tube center ($i = 1$), the condition $\frac{\partial\theta}{\partial\eta} = 0$ holds. Using the assumed ghost-node technique $\theta_{-1} = \theta_1$, the discretized equation for the first node is modified as follows:

$$(u_1 + 2r)\theta_1^{j+1} - 2r\theta_2^{j+1} = u_1\theta_1^j$$

This relation is implemented in the code as follows:

```
A(1,1) = u(1) + 2*r;
A(1,2) = -2*r;
```

The coefficient $-2 * r$ results from combining the coefficients of the left and right neighboring nodes because of symmetry.

c) Robin boundary condition at the wall ($\boldsymbol{Robin\ BC\ at\ \eta = 1}$)

At the wall ($i = N$), the boundary condition is $\frac{\partial \theta}{\partial \eta} + Bi \cdot \theta = 0$. In this code, a first-order backward difference is used to discretize this condition:

$$\frac{\theta_N - \theta_{N-1}}{\Delta\eta} + Bi \cdot \theta_N = 0 \quad \Rightarrow \quad (1 + Bi\Delta\eta)\theta_N - \theta_{N-1} = 0$$

This algebraic equation is placed directly in the last row of matrix A:

$A(N, N-1) = -1;$

$A(N, N) = 1 + Bi * d_eta;$

An important point is that the last row of matrix A does not correspond to a balance equation; rather, it represents the boundary-condition equation. Therefore, the corresponding value in the right-hand-side vector ($B(N)$) is set equal to zero.

4-1-3 System Solution and Time Marching

In the main loop $for\ j = 1: M-1$, the following operations are repeated:

1. **Update of the known vector (B): it is calculated from the temperature profile at the previous step** (θ^j) and the velocity profile.
2. **Solution of the equation system:** the MATLAB command `Theta_new = A \ B` uses Gaussian elimination (or the Thomas algorithm for tridiagonal matrices) to obtain the temperature profile at the new step (θ^{j+1}).

4-1-4 Analysis of Results and Plots (Visualization)

The presented code produces three types of graphical output that illustrate the physical behavior of the system:

1- Temperature-distribution contour (Figure 1):

A color map of temperature over the two-dimensional domain (ξ, η). This plot shows how the cold fluid ($T = 0$) is heated near the wall and how the thermal boundary layer grows along the tube.

2- Temperature profiles at different cross-sections (Figure 2):

This plot shows the variation of temperature with radius (η) at different axial locations (ξ).

- At the tube inlet ($\xi \approx 0$): the temperature profile is flat (value 1).
- At the tube outlet ($\xi = L$): the temperature decreases near the wall, and the profile tends toward a fully developed distribution.

3- Centerline and wall temperature variations (Figure 3):

- Blue line (Centerline): the temperature at the tube center, which changes later than all other points because of its distance from the wall.
- Red dashed line (Wall): the fluid temperature in contact with the wall. Because of the Robin condition (external convection), the wall temperature is not constant and is a function of the Biot number and the fluid temperature. The decrease in wall temperature along the tube indicates physically correct behavior (heat exchange with the colder external environment).

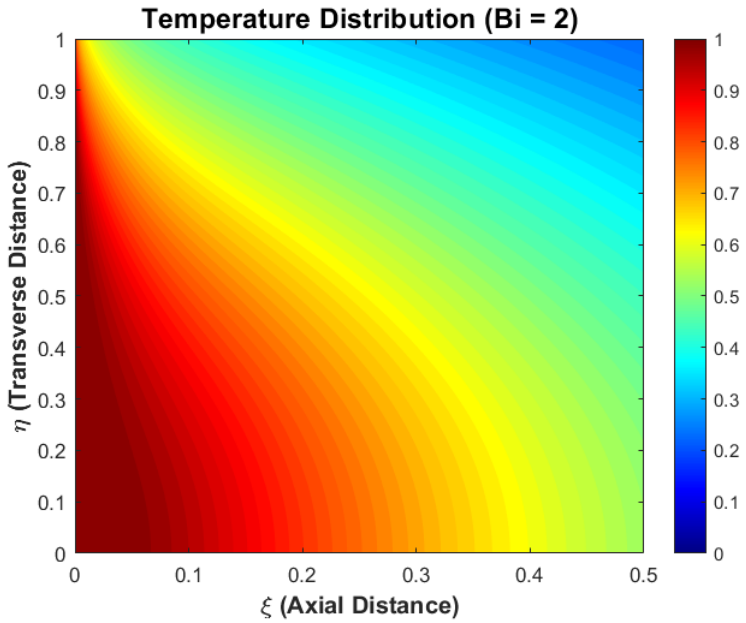


Figure 1: Color map of temperature over the two-dimensional domain(ξ, η).

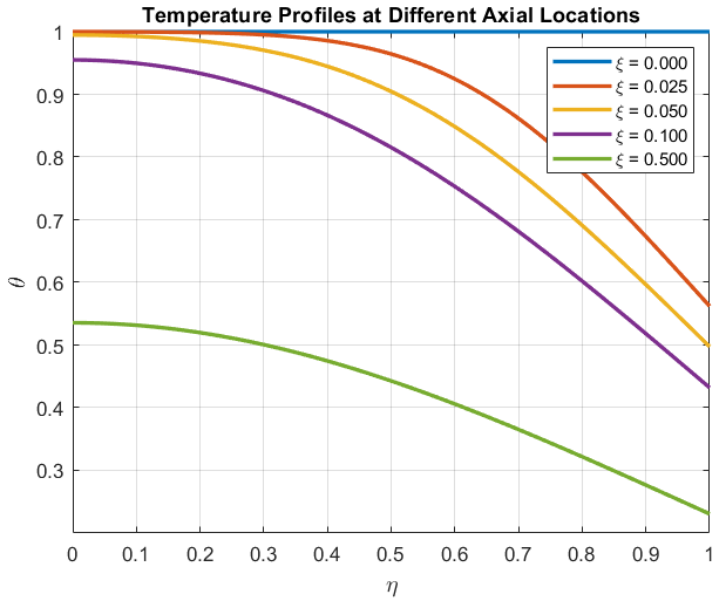


Figure 2: Temperature variation with radius (η) at different axial locations (ξ)

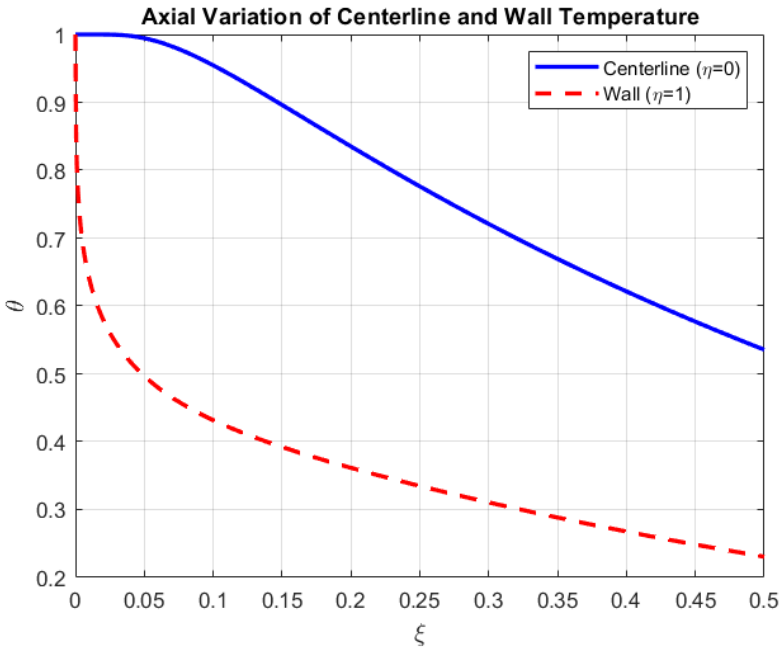


Figure 3: Variations of centerline and wall temperature

Stability and Convergence Assessment

4-2 Part Two: Validation of the Results

This code consists of three main parts: the numerical solution (FDM), the analytical solution, and comparison and plotting. The objective is to demonstrate that the results obtained by discretizing the differential equations are in complete agreement with the exact mathematical solution obtained from the power series.

4-2-1 Part One: Numerical Solution (*Implicit FDM*)

In the initial lines of the code (Part 1), the same implicit finite-difference method developed previously is implemented. The main features of this part are as follows:

1- Coefficient-matrix assembly (A):

1. Because the velocity profile $u(\eta)$ depends only on the transverse coordinate and does not change with time, the coefficient matrix A is a constant tridiagonal matrix. It is assembled only once at the beginning of the code to improve computational efficiency.
2. **Application of the symmetry condition: in the first row of the matrix** ($i = 1$), the ghost-node technique is used. By assuming $\theta_{-1} = \theta_{1}$,, a zero derivative at the channel centerline is ensured. This results in the coefficient $-2 * r$ in matrix entry (1,2).
3. **Application of the Robin boundary condition: in the last row** ($i = N$), the condition $Y'(1) + Bi \cdot Y(1) = 0$ is discretized using a first-order backward difference. This directly relates the wall temperature to the temperature at the penultimate node ($N - 1$):

$$(1 + Bi\Delta\eta)\theta_N - \theta_{N-1} = 0$$

This equation is placed in the last row of matrix A.

2- Solution of the system of equations:

- Within the marching loop, at each step the linear system $A\theta^{j+1} = B$ is solved. The right-hand-side vector B contains temperature information from the previous step.

4-2-2 Part Two: Analytical Solution

This part ($Part\ 2$) is the most complex portion of the code and implements the derived mathematical theory (separation of variables and the power-series method).

1- Determination of the eigenvalues

- **Challenge: the eigenvalues** (λ_n) are the roots of a transcendental characteristic equation that has no closed-form solution.
- **Code approach:** the MATLAB function `fzero` is used. This function searches for zeros of `char_eq`.
- **Initial guess: because the eigenvalues usually occur sequentially, after each root is found, the initial guess for the next root** ($guess$) is shifted forward to avoid finding repeated roots.

2- Calculation of the eigenfunctions using the power series

The core of the analytical solution is the local function `power_series_val_deriv` at the end of the code.

This function implements the recurrence relation derived in the hand calculation:

$$b_k = \frac{\lambda^2(b_{k-2} - b_{k-1})}{2k(2k-1)}$$

- **Calculation loop: the function evaluates up to 40 terms of the series (or until the terms become**

negligible).

- **Dual output:** the function returns both the value Y(η) and its derivative Y′(η), because both are required to check the Robin boundary condition Y′(1) + Bi Y(1) = 0.

3- Calculation of the expansion coefficients (Coefficients Cn)

- According to Sturm–Liouville theory, the final solution is a linear combination of the eigenfunctions : $\theta = \sum C_n Y_n$.
- The coefficient C_n is calculated from the orthogonality relation:

$$C_n = \frac{\int_0^1 (1-\eta^2) Y_n(\eta) \cdot 1 \, d\eta}{\int_0^1 (1-\eta^2) Y_n^2(\eta) \, d\eta}$$

- In the code, these integrals are evaluated using the trapz command (numerical trapezoidal rule) over the grid points.

4- Construction of the final series

- Finally, the analytical solution is constructed by summing the first five modes (Modes 1 to 5). Experience indicates that, for heat-transfer problems, the first five terms provide very high accuracy (greater than 99%).

4-2-3 Part Three: Validation

This section ($Part\ 3$) is responsible for displaying the results.

Selection of sections: three different axial sections (the beginning, middle, and end of the tube) are selected to verify the solution throughout the entire domain.

Display method:

- Solid lines ($Solid\ Lines$): represent the analytical (exact) solution.
- Markers ($Symbols$ - circles/squares): represent the numerical solution.
- Interpretation of the plot: as observed in the output, the numerical-solution markers lie directly on the analytical-solution curves. This means that the numerical error is very small and that the implicit finite-difference code has been implemented correctly.

4-2-4 Helper Functions (*Helper Functions*) at the End of the Code

1- char_eq(lam, Bi):

This function defines the characteristic equation. An eigenvalue λ is valid when the output of this function is zero.

Input: a guessed value of λ.

Operation: evaluates Y(1) and Y'(1) using the power series and computes the characteristic-equation residual.

Output: residual of the equation $Y'(1) + Bi \cdot Y(1)$.

2- power_series_val_deriv(lam, eta):

This is the most critical function; it evaluates the power series for a specified η and λ.

- It uses the array b to store the series coefficients.
- It includes a stopping condition (break) so that, if the series terms become very small (less than10^{-12}), the loop terminates and the computation is accelerated.

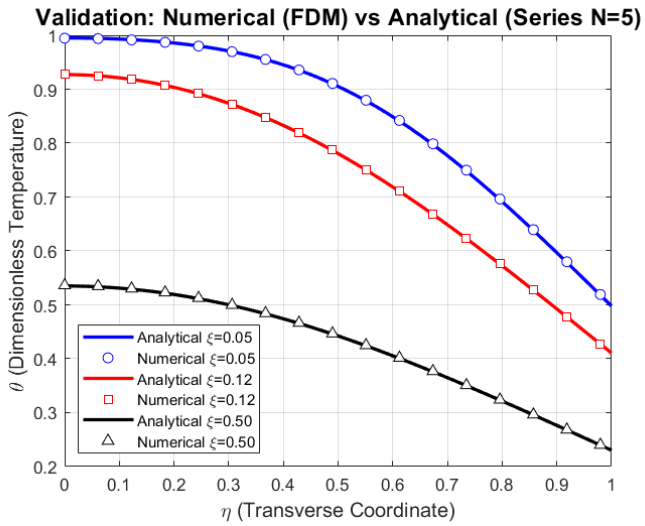


Figure 4: Output of the second part of the MATLAB code for validating the numerical results against the analytical solution

Appendix 1: MATLAB Code Implemented for the Numerical Solution

```
clc; clear; close all;


% 1. Parameters Definition
Bi = 2.0;
L_xi = 0.5;
N = 100;
M = 1000;


d_eta = 1.0 / (N - 1);
d_xi = L_xi / M;


% Grid Initialization
eta = linspace(0, 1, N);
xi = linspace(0, L_xi, M);


u = 1 - eta.^2;


% Stability parameter r = d_xi / d_eta^2
r = d_xi / (d_eta^2);


% 2. Initialization
Theta = zeros(N, M);


% Initial Condition: Theta = 1 at xi = 0 (Uniform Inlet Temperature)
```

```
Theta(:, 1) = 1.0;
A = zeros(N, N);

% 3-1. Symmetry Boundary at eta = 0 (Node i=1)
% Equation: (u_1 + 2r)*T_1 - 2r*T_2 = u_1*T_old
A(1, 1) = u(1) + 2*r;
A(1, 2) = -2*r;


for i = 2:N-1
   A(i, i-1) = -r;
   A(i, i)   = u(i) + 2*r;
   A(i, i+1) = -r;
end


A(N, N-1) = -1;
A(N, N)   = 1 + Bi * d_eta;


B = zeros(N, 1);
for j = 1:M-1
   % Construct RHS Vector
   % For interior and symmetry: B(i) = u(i) * Theta(i, j)
   B(1:N-1) = u(1:N-1)' .* Theta(1:N-1, j);

   % For Wall BC: RHS is 0 (homogeneous BC)
   B(N) = 0;


   Theta_new = A \ B;

   % Update solution
   Theta(:, j+1) = Theta_new;
end


%% 5. Visualization and Results
% Contour Plot
figure(1);
[Xi_grid, Eta_grid] = meshgrid(xi, eta);
contourf(Xi_grid, Eta_grid, Theta, 50, 'LineColor', 'none');
colorbar;
colormap('jet');
caxis([0 1]);
xlabel('\xi (Axial Distance)', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('\eta (Transverse Distance)', 'FontSize', 12, 'FontWeight', 'bold');
title(['Temperature Distribution (Bi = ', num2str(Bi), ')'], 'FontSize', 14);
   set(gca, 'YDir', 'normal'); % Ensure y-axis starts from 0 at bottom


% Temperature Profiles at specific axial locations
figure(2);
hold on;
```

```
plot_indices = [1, round(M/20), round(M/10), round(M/5), M];
legends = {};
for idx = plot_indices
    plot(eta, Theta(:, idx), 'LineWidth', 2);
    legends{end+1} = ['\xi = ', num2str(xi(idx), '%.3f')];
end
xlabel('\eta', 'FontSize', 12);
ylabel('\theta', 'FontSize', 12);
title('Temperature Profiles at Different Axial Locations');
legend(legends);
grid on;
box on;
% Centerline and Wall Temperature
figure(3);
plot(xi, Theta(1, :), 'b-', 'LineWidth', 2); hold on;
plot(xi, Theta(N, :), 'r--', 'LineWidth', 2);
xlabel('\xi', 'FontSize', 12);
ylabel('\theta', 'FontSize', 12);
legend('Centerline (\eta=0)', 'Wall (\eta=1)');
title('Axial Variation of Centerline and Wall Temperature');
grid on;
```

Appendix 2: MATLAB Code Implemented for Validation of the Numerical Results against the Analytical Solution

```
clc; clear; close all;


%% --- PART 1: NUMERICAL SOLUTION (Implicit FDM) ---
fprintf('Running Numerical Solution...\n');


% 1. Parameters
Bi = 2.0;           % Biot Number
L_xi = 0.5;          % Dimensionless length
N = 50;             % Grid points in eta (kept low for visibility of markers)
M = 500;             % Steps in xi


d_eta = 1.0 / (N - 1);
d_xi = L_xi / M;
eta = linspace(0, 1, N);
xi = linspace(0, L_xi, M);
u = 1 - eta.^2;       % Velocity profile
r = d_xi / (d_eta^2);   % Stability parameter


% 2. Initialization
Theta_Num = zeros(N, M);
Theta_Num(:, 1) = 1.0;  % IC
```

```
% 3. Matrix Assembly
A = zeros(N, N);
% Symmetry (i=1)
A(1, 1) = u(1) + 2*r;
A(1, 2) = -2*r;
% Interior
for i = 2:N-1
   A(i, i-1) = -r;
   A(i, i)   = u(i) + 2*r;
   A(i, i+1) = -r;
end
% Wall (Robin BC)
A(N, N-1) = -1;
A(N, N)   = 1 + Bi * d_eta;


% 4. Time Stepping
B_vec = zeros(N, 1);
for j = 1:M-1
   B_vec(1:N-1) = u(1:N-1)' .* Theta_Num(1:N-1, j);
   B_vec(N) = 0;
   Theta_Num(:, j+1) = A \ B_vec;
end


fprintf('Running Analytical Solution...\n');


NumModes = 5;
Lambdas = zeros(NumModes, 1);
Cn = zeros(NumModes, 1);


options = optimset('TolX', 1e-6, 'Display', 'off');
guess = 2; % Initial guess
for n = 1:NumModes
   Lambdas(n) = fzero(@(lam) char_eq(lam, Bi), guess, options);
   guess = Lambdas(n) + 3;
end


fprintf('Eigenvalues found: \n');
disp(Lambdas);
Yi = zeros(N, NumModes);


for n = 1:NumModes
   lam = Lambdas(n);

   Y_vals = get_eigenfunction(lam, eta);
   Yi(:, n) = Y_vals;

   % Numerical Integration (Trapezoidal rule)
   Numerator = trapz(eta, (1 - eta.^2) .* Y_vals);
```

```
    Denominator = trapz(eta, (1 - eta.^2) .* Y_vals.^2);
    Cn(n) = Numerator / Denominator;
end

Comparison_Indices = [round(M/10), round(M/4), M]; % Locations to compare
Theta_Ana = zeros(N, length(Comparison_Indices));


for k = 1:length(Comparison_Indices)
    idx_xi = Comparison_Indices(k);
    xi_val = xi(idx_xi);

    T_sum = zeros(1, N);
    for n = 1:NumModes
        term = Cn(n) * exp(-Lambdas(n)^2 * xi_val) * Yi(:, n)';
        T_sum = T_sum + term;
    end
    Theta_Ana(:, k) = T_sum;
end




figure('Name', 'Validation of Numerical vs Analytical', 'Color', 'w');
hold on;
colors = {'b', 'r', 'k'};
markers = {'o', 's', '^'};

for k = 1:length(Comparison_Indices)
    idx_xi = Comparison_Indices(k);
    xi_current = xi(idx_xi);

    plot(eta, Theta_Ana(:, k), [colors{k} '-'], 'LineWidth', 2, ...
        'DisplayName', ['Analytical \xi=' num2str(xi_current, '%.2f')]);

    plot(eta(1:3:end), Theta_Num(1:3:end, idx_xi), [colors{k} markers{k}], ...
        'MarkerFaceColor', 'w', 'DisplayName', ['Numerical \xi=' num2str(xi_current, '%.2f')]);
end


xlabel('\eta (Transverse Coordinate)', 'FontSize', 12);
ylabel('\theta (Dimensionless Temperature)', 'FontSize', 12);
title(['Validation: Numerical (FDM) vs Analytical (Series N=', num2str(NumModes), ')'], 'FontSize', 14);
legend('Location', 'SouthWest');
grid on;
box on;


function res = char_eq(lam, Bi)
    [Y1, Yp1] = power_series_val_deriv(lam, 1);
    res = Yp1 + Bi * Y1;
end
```

```
function Y_vals = get_eigenfunction(lam, eta_vec)
    Y_vals = zeros(size(eta_vec));
    for i = 1:length(eta_vec)
        [Y_vals(i), ~] = power_series_val_deriv(lam, eta_vec(i));
    end
end

function [val, deriv] = power_series_val_deriv(lam, eta)
    n_terms = 40;
    b = zeros(n_terms, 1);
    b(1) = 1; % b0 corresponds to index 1

    val = b(1);      % Y = b0 * eta^0
    deriv = 0;       % Y' = 0

    if n_terms > 1
        b(2) = -lam^2 / 2 * b(1);
        term = b(2) * eta^2;
        val = val + term;
        deriv = deriv + 2 * b(2) * eta;
    end



    for k = 2:n_terms-1

        numerator = lam^2 * (b(k-1) - b(k));
        denominator = 2*k * (2*k - 1);

        b(k+1) = numerator / denominator;

        term = b(k+1) * eta^(2*k);
        term_deriv = 2*k * b(k+1) * eta^(2*k-1);

        val = val + term;
        deriv = deriv + term_deriv;

        % Check convergence
        if abs(term) < 1e-12
            break;
        end
    end
end
```